

ETZIKOM AGCM, AB, Canada

WMO: 714090

Lat: 49.5528N

Lon: 111.0542W

Elev: 845

StdP: 91.58

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.2	-26.2	-33.4	0.2	-30.1	-29.4	0.3	-26.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	32.7	16.2	30.7	16.0	28.6	15.5	18.5	26.7	17.5	26.5	16.6	25.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	12.4	21.4	14.4	11.4	20.2	13.4	10.6	19.1	55.9	26.7	52.5	26.6	49.8	25.9	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-35.6	36.3	2.8	1.6	-37.6	37.5	-39.2	38.5	-40.8	39.4	-42.8	40.6		
(5)			-35.7	20.6	2.7	1.5	-37.7	21.6	-39.3	22.5	-40.8	23.3	-42.8	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-7.7	-8.8	-1.3	5.3	11.1	15.3	19.0	18.4	13.3	5.7	-1.6	-8.6		
	DBStd	11.80	8.44	8.75	7.76	5.03	4.40	3.36	3.03	3.59	4.81	5.32	7.73	8.42		
	HDD10.0	2717	548	527	350	153	40	2	0	0	20	151	348	578		
	HDD18.3	4952	807	761	608	391	227	102	28	44	161	392	597	836		
	CDD10.0	927	0	0	1	12	74	162	278	261	121	17	2	0		
	CDD18.3	120	0	0	0	0	2	12	48	47	12	0	0	0		
	CDH23.3	2385	0	0	0	8	104	248	875	851	288	10	0	0		
	CDH26.7	884	0	0	0	1	19	65	337	350	112	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	364	17	13	19	30	47	83	36	34	31	21	15	14		
	PrecMax	532	44	29	50	68	102	207	125	80	68	70	32	32		
	PrecMin	193	6	2	1	7	7	15	3	2	0	6	2	0		
	PrecStd	76	10	7	12	15	24	41	28	21	20	14	9	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	10.6	18.0	24.1	29.0	32.4	34.8	34.7	33.1	25.0	17.7	9.4		
		MCWB	5.1	5.4	8.6	11.5	14.0	17.2	16.6	16.4	15.5	13.3	8.7	4.3		
	2%	DB	7.3	7.1	14.8	20.8	26.3	28.3	32.5	32.6	30.3	20.7	13.5	6.1		
		MCWB	3.0	3.1	7.2	10.1	13.1	15.8	16.8	15.8	14.9	11.0	6.9	2.7		
	5%	DB	5.2	4.7	11.9	17.5	23.7	26.1	30.4	30.7	26.7	17.6	10.7	4.4		
		MCWB	2.1	1.8	5.7	8.1	12.1	15.0	16.7	15.5	13.8	9.6	5.6	1.7		
	10%	DB	3.7	2.9	8.8	14.8	21.0	24.0	28.3	28.3	23.5	14.9	8.0	2.8		
		MCWB	1.3	0.7	3.9	6.7	11.0	14.1	16.2	15.3	12.5	7.9	3.8	0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	6.0	9.0	12.0	14.8	19.3	20.6	18.6	17.2	13.7	9.5	5.1		
		MCDB	9.5	10.0	17.0	22.1	26.0	26.2	29.0	27.7	27.2	23.4	17.4	8.4		
	2%	WB	3.3	3.4	7.5	10.4	13.7	17.3	19.1	17.6	15.6	11.7	7.5	3.0		
		MCDB	6.6	6.5	13.8	20.0	23.8	26.3	26.6	27.5	26.9	18.8	12.6	5.8		
	5%	WB	2.3	1.9	6.0	8.6	12.8	15.9	18.1	16.8	14.3	9.9	5.8	1.8		
		MCDB	4.9	4.5	11.5	16.5	21.8	24.1	26.5	27.0	24.5	16.3	10.3	4.3		
	10%	WB	1.3	0.7	4.1	7.0	11.8	14.9	17.2	16.0	13.1	8.5	4.1	0.4		
		MCDB	3.8	2.8	8.4	14.0	19.4	22.5	25.5	25.7	23.2	14.4	7.5	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.0	11.6	13.4	14.1	13.8	16.2	16.3	15.5	13.6	11.5	10.2		
		MCDBR	10.2	12.9	15.5	19.0	19.1	17.8	20.0	20.9	21.3	17.9	14.5	11.5		
		MCWBR	7.5	9.5	9.0	9.6	8.1	7.8	7.6	7.5	8.5	9.2	8.7	8.5		
		MCDBR	9.9	12.4	15.2	18.0	16.5	16.2	16.3	17.1	19.5	16.8	13.7	11.5		
	5% WB	MCWBR	7.6	9.3	9.1	9.3	7.4	7.7	7.4	6.6	8.0	9.0	8.6	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.269	0.293	0.343	0.364	0.356	0.355	0.376	0.333	0.295	0.265	0.254		
	taud		2.418	2.470	2.467	2.324	2.311	2.387	2.402	2.314	2.411	2.504	2.496	2.417		
	Ebn at Noon		804	897	927	902	894	902	897	857	865	850	805	768		
	Edh at Noon		62	75	90	116	123	115	111	116	94	72	57	53		
(44) All-Sky Solar Radiation	RadAvg		1.28	2.28	3.58	4.94	5.84	6.48	7.05	5.66	4.02	2.52	1.42	1.00		
	RadStd		0.11	0.18	0.25	0.34	0.43	0.38	0.36	0.36	0.33	0.24	0.09	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page